

BREAK INTO AEROSPACE

Kongsberg Gruppen

Engineering Interview Preparation Guide

Targeted for the 2025–2026 Recruitment Cycle

Kongsberg Gruppen is one of the most distinctive companies in this series — a Norwegian state-majority-owned technology group operating at the intersection of precision missiles, maritime autonomy, and space systems, whose products are deployed in NATO member states across four continents and whose defence business is in extraordinary growth as Europe rearranges its security architecture after 2022. This guide consolidates the preparation tasks, signal areas, and strategic context needed to compete effectively across all relevant engineering roles.

PART I · WHAT KONGSBERG ACTUALLY LOOKS FOR

By Seniority Level

- **Intern / Summer Student** — Kongsberg’s 2025 summer internship programme received 2,800 applications — a 44% year-on-year increase and the highest ever — with 151 students joining in 2024 across all business areas. The selection process is application-plus-interview with no aptitude test. Internships are offered in two formats: embedded department placements and multidisciplinary summer projects, primarily intended for students in science and technology. The multidisciplinary projects — such as Project Revolution Green (autonomous vessel simulation) and Project Summit2028 (sailing technology for the Olympics) — are high-visibility, executive-facing, and genuinely challenging. Students present results to executive management at the end of summer.

What Kongsberg screens for at intern level is intellectual curiosity and technical engagement rather than a specific credential profile. Kongsberg emphasises collaboration and inclusivity — working closely with colleagues, customers, suppliers, and partners worldwide — and the spirit of “ONE KONGSBERG, where we make the impossible possible by performing together.” Interns who demonstrate cross-discipline curiosity rather than narrow specialisation are more strongly selected.

- **Mid-Level (Engineer, Senior Engineer, ~2–7 years)** — Kongsberg’s mid-career interview process typically involves a Teams screen followed by behavioural testing and then an interview with managers and HR — described as professional, to the point, and with a strong focus on technical aspects alongside behavioural questions structured around “tell us about a time you...”. For domain-specific roles, the technical probing is direct and specific: a maritime engineering role interview asked candidates to name components from images, name the lubrication fluid shown, and describe a manoeuvring thruster schematic. This product-specific technical knowledge requirement is not softened by the conversational tone of the interview.

The specific context driving mid-level hiring in 2025–2026: Kongsberg Defence & Aerospace (KDA) is expanding rapidly across all product lines driven by European defence budget increases following Russia’s invasion of Ukraine. KDA is hiring approximately 30 engineers in Fredrikstad, establishing a presence in the city for the first time, alongside ongoing expansion at its Kongsberg headquarters. NSM deliveries to 14 nations, NASAMS expansions, JSM deliveries to Japan and Australia, Protector remote weapon station contracts for Poland and Germany, and a new F-35 components agreement with

Lockheed Martin worth NOK 2 billion are all simultaneously driving headcount. Engineers who join Kongsberg now are joining an organisation in genuine acceleration.

- **Senior (Principal Engineer, Technical Lead, Fellow)** — At senior level, Kongsberg looks for engineers who can carry technical authority across complex multi-nation programme deliveries, who can represent Kongsberg's technical position to NATO customer programme offices, and who can participate in the kind of cross-organisational collaboration that NASAMS (co-developed with Raytheon), JSM (integrated into the F-35), and the 3SM Tyrving supersonic missile (a consortium with Diehl, MBDA, and Nammo) require. Senior engineers at Kongsberg are expected to own technical threads across hardware, software, and systems — and the engineering leadership is described as listening and responsive.

The senior differentiator at Kongsberg is the ability to operate at the intersection of Norwegian defence requirements and international alliance architecture — understanding not just what a system must do technically but how it must integrate with NATO's Integrated Air and Missile Defence (IAMD) framework, Link 16, and the interoperability protocols that govern multinational force operations.

Hands-On Signal Areas

Domain	Core Competencies Expected
Missile Systems Engineering	NSM and JSM are Kongsberg's highest-profile defence products. GNC, passive IR seeker design, terminal homing algorithms, composite stealth airframe structures, and warhead fuzing are all active engineering domains at KDA. Engineers with missile systems experience from any Western nation are directly relevant.
Air Defence Systems Engineering	NASAMS is a medium-range surface-to-air missile system co-developed by Kongsberg and Raytheon, used by 13 NATO nations. Engineers with radar integration, C2 systems, or air defence architecture experience are in persistent demand.
Maritime Systems	Kongsberg Maritime is a world leader in dynamic positioning systems, autonomous vessel control, and underwater sensors. Engineers with DP system design, vessel automation, or hydrodynamic simulation backgrounds are specifically relevant.
Software Engineering (C++ / Real-Time Embedded)	Kongsberg's guidance systems, fire control systems, and maritime automation all run on safety-critical embedded software. Software roles emphasise QA methodology and past experience with quality processes as specific interview topics.
Norwegian Language Fluency	For Kongsberg-based roles at headquarters in Kongsberg, Horten, and Kjeller — where the majority of KDA engineering is located — Norwegian is the working language of daily engineering. English is widely used in international programme contexts but team discussions, design reviews, and informal collaboration all happen in Norwegian.
Norwegian / NATO-Nation Security Clearance	KDA's classified defence work requires Norwegian security clearance (<i>sikkerhetsgodkjenning</i>) at the appropriate level, governed by the Norwegian National Security Authority (NSM). Norwegian citizenship is the strongest credential; EU/EEA nationals with significant Norwegian residency can often be sponsored.
Systems Engineering / MBSE	Kongsberg's complex multi-subsystem products require rigorous systems engineering. Engineers with MBSE experience or strong systems integration backgrounds are broadly relevant across all business areas.

How Kongsberg's Interview Format Differs from Standard Tech and Other Companies

- **The process is lean and direct by European defence standards.** The typical Kongsberg process involves an online application, behavioural testing, and one or two interview rounds with managers and HR — averaging about 25 days from application to decision. There is no large assessment centre, no multi-day onsite, and no formal aptitude test. The process is shorter and less formal than BAE Systems, Airbus, or Leonardo.

- **Technical probing is product-specific and direct.** Kongsberg interviewers are engineers working on specific product lines. For maritime roles, component-identification questions and system schematic comprehension are used directly. For defence roles, familiarity with the specific weapons system or platform is expected. This is not a generic engineering competency test — it is a domain-knowledge verification.
- **Behavioural questions use standard “tell me about a time you...” format.** Alongside technical questions, interviews use behavioural questions covering QA methods, past challenges, and how processes helped resolve problems. The most common themes cluster around quality, cross-team collaboration, operating in ambiguous technical situations, and customer interface.
- **English is broadly acceptable for international applicants but Norwegian is the working language domestically.** English-language interviews are common for international candidates — particularly for roles at international subsidiaries. However, for roles at Kongsberg HQ and KDA's main engineering sites, Norwegian fluency is a practical daily requirement.
- **The “ONE KONGSBERG” culture has practical interview implications.** Questions about how you have worked across organisational boundaries, contributed to team outcomes rather than individual achievements, and operated in complex multi-stakeholder environments are live at Kongsberg in ways that reflect its genuinely integrated multi-business structure.
- **Norwegian security clearance eligibility is screened early.** For KDA roles involving classified defence work, clearance eligibility is verified before interviews proceed far. Norwegian citizenship is the strongest credential. Non-EEA candidates face material access restrictions on classified KDA work — though unclassified roles in maritime and digital remain accessible.

Honest caveat on format variation: Kongsberg Gruppen's four business areas — Kongsberg Defence & Aerospace, Kongsberg Maritime, Kongsberg Discovery, and Kongsberg Digital — have genuinely different cultures. KDA at Kongsberg HQ is defence-security focused, clearance-governed, and programme-oriented. Kongsberg Maritime at Horten, Ålesund, and 35 international locations is commercially focused, globally mobile, and customer-service oriented. Kongsberg Digital at Oslo is a software and digital twin company with a more technology-startup feel. The interview format and technical culture differ significantly between these. Filter Glassdoor reviews by business area wherever possible.

PART II · 20 SPECIFIC PREPARATION TASKS

Intern / Summer Student

1. Apply to the Summer Project Track, Not Just Department Placements — and Research the Specific Project Before Applying 2–3 h

Skill: Application strategy that maximises selection probability and programme value

Objective: Summer projects at Kongsberg — such as Project Revolution Green (autonomous vessel simulation) and Project Summit2028 (sailing technology for the Olympics) — involve multidisciplinary teams presenting results to executive management at summer's end. These are high-visibility, technically substantive, and are explicitly designed to identify future full-time employees. Candidates who research the specific project's goals, understand what has been built in previous summers, and can articulate a specific contribution they would make are more compelling than generic applicants.

Done Signal: You have read the publicly available description of your target summer project (Revo-

lution Green, Summit2028, or equivalent), can explain what was built by previous cohorts and what the 2026 team is expected to achieve, and have written a cover letter that specifically names your technical contribution to the project — not to Kongsberg generically. 2–3 h

2. Prepare a Cover Letter That Names a Specific Kongsberg Product or Technology — and Your Technical Connection to It 2 h

Skill: Written application differentiation in a high-volume selection process

Objective: Kongsberg received 2,800 intern applications for 2025 — a record. Selection from this volume requires differentiation at the document screening stage. A cover letter that names a specific Kongsberg product (NSM, NASAMS, dynamic positioning system, autonomous vessel control) and explains one technically specific reason it interests you is immediately differentiated from the hundreds of letters expressing generic enthusiasm for “innovative technology in defence and maritime.”

Done Signal: You have a one-page cover letter that names the specific Kongsberg project or business area you are applying to, identifies one specific technical product or programme within it, explains what specifically interests you about one engineering challenge it presents, and connects your academic background to a concrete contribution you could make.

3. Develop Working Knowledge of the “ONE KONGSBERG” Structure — Four Business Areas, Their Products, and How They Collaborate 3 h

Skill: Corporate structure literacy that enables a peer conversation with interviewers from any business area

Objective: Kongsberg’s “ONE KONGSBERG” identity emphasises cross-business collaboration as the source of competitive advantage — making the impossible possible by performing together. Interns who understand that Kongsberg Maritime’s dynamic positioning technology informs KDA’s naval vessel systems, or that Kongsberg Digital’s simulation tools are used to develop maritime autonomy, are demonstrating institutional engagement that distinguishes them from those who know only the business area they applied to.

Done Signal: For each of the four business areas — Kongsberg Defence & Aerospace, Kongsberg Maritime, Kongsberg Discovery, and Kongsberg Digital — you can state the primary product families, the primary customer types, and one specific example of how this business area’s technology connects to another business area’s work. You can do this conversationally, without a list.

4. Prepare Behavioural Answers in the “Tell Me About a Time You...” Format — Focusing on Collaboration and Quality 2–3 h

Skill: Behavioural interview preparation for Kongsberg’s standard interview format

Objective: Kongsberg behavioural questions specifically probe QA methods — what quality processes you have used, what problems they helped resolve — alongside standard collaboration and challenge-navigation themes. For an intern, these stories come from academic projects, lab work, or extracurricular activities. The quality-process dimension is specific to Kongsberg’s culture: a company that builds missiles, precision instruments, and safety-critical maritime automation treats quality as an engineering discipline, not a department.

Done Signal: You have four STAR stories covering: a technical problem you solved collaboratively, a quality or accuracy issue you identified and addressed, a time you worked across disciplinary boundaries, and a failure you learned from. Each ends with a specific outcome. At least one references a quality or verification step you personally implemented.

5. Assess Norwegian Language Level and Begin Targeted Improvement If Below B1 for Kongsberg-Based Roles 1 h + ongoing

Skill: Working language readiness for a Norwegian-headquartered engineering environment

Objective: Kongsberg's main engineering sites — Kongsberg city (KDA), Horten (maritime), Kjeller (aerospace) — operate in Norwegian. Design meetings, informal technical discussions, and daily team collaboration happen in Norwegian. English is widely used for international programme interface and in international subsidiaries, but candidates who can participate in Norwegian, even at B1 level, integrate faster and contribute sooner.

Done Signal: You have an honest, calibrated assessment of your Norwegian level (none / A1–A2 / B1 / B2 / C1) and — if below B1 — a concrete improvement plan with a target date. You can introduce yourself and describe your technical background in Norwegian, even if you need English for technical content.

6. Understand Kongsberg's Security Clearance Requirements and Your Personal Eligibility 1–2 h

Skill: Clearance eligibility self-assessment before applying to KDA roles

Objective: KDA's classified missile and air defence programmes require Norwegian security clearance (*sikkerhetsgodkjenning*) at levels ranging from KONFIDENSIELT to STRENGT HEMMELIG. Norwegian citizenship is the broadest access. EU/EEA nationals with significant Norwegian residency can typically be sponsored; non-EEA nationals face material access restrictions on classified work. Understanding your eligibility before applying to specific KDA roles prevents wasted effort and recruiter friction.

Done Signal: You know your citizenship and residency status, have identified which KDA roles and product areas are accessible given your nationality, and can answer the recruiter's clearance eligibility question clearly and without hesitation. If your nationality creates restrictions, you know which Kongsberg Maritime, Discovery, or Digital roles remain open to you.

Mid-Level (~2–7 years)

All intern tasks still apply. Behavioural stories must come from professional contexts. Norwegian language must be working level for Kongsberg-based roles. Security clearance eligibility must be verified before applying.

7. Build Working Technical Knowledge of the NSM and JSM — Architecture, Seeker, Terminal Guidance, and Export Reach 4–5 h

Skill: KDA's flagship products at the technical depth the interview expects

Objective: The Naval Strike Missile is a fifth-generation stealth cruise missile selected by 14 nations, designed for littoral and open-sea use with a sea-skimming flight profile and random terminal manoeuvres to defeat enemy countermeasures. The JSM is an air-launched variant designed to fit the F-35's internal weapons bay. For mid-level engineers interviewing for missile systems, GNC, seekers, or combat systems roles at KDA, understanding the NSM/JSM architecture at technical depth is the baseline expected knowledge. Candidates who know only that the NSM is a Norwegian missile that goes fast are demonstrating engagement at marketing-brochure depth.

Done Signal: You can explain what a passive IR seeker is and why it is preferred over active radar for survivability, describe the sea-skimming flight profile and what guidance law considerations it requires, explain why composite materials contribute to low radar cross-section, and describe what "terminal manoeuvres" require of the guidance and control system. You can answer one follow-up

question on each.

4–5 h

8. Study NASAMS — Its Architecture, the AMRAAM Integration, and the Kongsberg/Raytheon Co-Development Model

4 h

Skill: KDA's most widely deployed air defence system at engineering depth

Objective: NASAMS is a medium-range surface-to-air missile system first developed by Kongsberg and Raytheon for the Norwegian Armed Forces and now used by 13 nations. Its architecture — integrating existing AIM-120 AMRAAM missiles with a Kongsberg-developed fire control system, multiple radar options, and C2 software — is the template for Kongsberg's approach to systems integration: taking best-in-class components and providing the integration, software, and operational interface that makes them work as a coherent defended system. Engineers who understand how the FDC (Fire Distribution Centre) communicates with radar inputs and launch platforms, and how the system connects to NATO IAMD networks via Link 16, are demonstrating the systems thinking KDA values.

Done Signal: You can explain the NASAMS system architecture (radar, FDC, launchers, missile), describe what the FDC does and how it differs from the radar system, explain how AMRAAM's active radar seeker interacts with the ground-based fire control after handover, and describe what NATO IAMD interoperability requires from the command and control layer. You can answer one follow-up question on each.

9. Prepare a Story About Working on a Safety-Critical or Mission-Critical Engineering System — Where Verification and Validation Rigour Mattered

2 h

Skill: V&V discipline for systems where failure has severe operational consequences

Objective: Kongsberg specifically probes QA methods and past experience with quality processes — asking what processes were used, what problems they helped resolve, and how quality affected the engineering outcome. For a company whose products either protect people from incoming missiles (NASAMS) or strike defended targets at standoff range (NSM/JSM), verification and validation are not optional extras — they are the engineering discipline that determines whether the system performs when it must. The story must go beyond “we followed the process” to “the process caught this specific thing that would have failed in this specific way.”

Done Signal: You have a story about V&V on a safety-critical or mission-critical system where the process identified a failure mode that would otherwise have reached the customer or been discovered operationally. The story names the specific V&V step (integration test, fault injection, environmental qualification, HIL simulation, formal review), the specific failure mode it caught, and the design or software change that resulted.

10. Understand the Norwegian Defence Organisation — Forsvarsmateriell (FMA), the Norwegian Armed Forces, and How They Interface with KDA

3–4 h

Skill: Norwegian defence acquisition framework operational literacy

Objective: Kongsberg Defence & Aerospace's primary national customer is the Norwegian Ministry of Defence, managed through Forsvarsmateriell (FMA — Norwegian Defence Materiel Agency). FMA is the Norwegian counterpart of ATLA in Japan, DES in the UK, or DAPA in Korea. Understanding how Norwegian defence procurement works — how requirements originate in the Norwegian Armed Forces (Forsvaret), flow through FMA for programme management, and are contracted to KDA — is the operational context for all KDA's national programme work. Mid-level engineers who understand this framework can engage meaningfully in programme discussions from their first

week.

3–4 h

Done Signal: You can explain what Forsvarsmateriell is and its role in Norwegian defence acquisition, describe how a Norwegian Armed Forces operational requirement becomes a KDA engineering specification, explain what *Materielldirektiv* (material directives) and *Forsvarets Materiellhåndbok* (military handbook) mean for KDA's engineering standards, and describe how KDA interfaces with FMA differently for national programmes versus export contracts.

11. Build Working Knowledge of Dynamic Positioning Systems for Kongsberg Maritime Applicants — How DP Works and What Kongsberg's Contribution Is

4 h

Skill: Maritime automation domain knowledge for KDA's largest business sibling

Objective: Kongsberg Maritime is one of the world's leading DP system providers, with automation, dynamic positioning, and navigation systems installed on over 30,000 vessels. Mid-level engineers interviewing for Kongsberg Maritime roles who cannot explain what a dynamic positioning system does — how it integrates GPS, acoustic positioning, wind sensors, and thruster control to hold station automatically — are demonstrating they have not engaged with the product they would be supporting or developing. This is one of the product-knowledge probes confirmed in interview accounts.

Done Signal: You can explain what dynamic positioning is and why it is needed (station-keeping without anchors, for drillships, cable layers, offshore wind installation vessels), describe the three DP classes (DP1, DP2, DP3) and the redundancy requirements for each, explain how a DP system uses position reference inputs and a Kalman filter, and describe what thruster allocation logic does. You can answer a follow-up question on failure mode management in a DP2 system.

12. Prepare a Story About Working Across Organisational or National Boundaries — Specifically Managing a Technical Interface with a Partner Organisation

1–2 h

Skill: Cross-organisational collaboration at the level Kongsberg's multinational programmes require

Objective: Kongsberg's most important products are co-developed or co-produced with international partners: NASAMS with Raytheon, JSM integration with Lockheed Martin (F-35), 3SM Tyrfing with Diehl/MBDA/Nammo, NSM manufacturing in Australia with Thales. Managing the technical interface with a partner organisation — aligning on interface specifications, resolving integration issues, maintaining configuration control across organisational boundaries — is a daily reality for KDA engineers. Mid-level engineers who have done this in practice are demonstrating readiness for Kongsberg's programme environment.

Done Signal: You have a story about a technical interface with an external partner organisation — you can name both organisations, describe the interface being managed (an ICD, a data exchange agreement, a joint test procedure), the technical tension that arose at the interface, how it was resolved, and the formal document that recorded the resolution.

13. Understand the Protector Remote Weapon Station — Its Variants, Customers, and What Systems Engineering It Involves

3 h

Skill: KDA's globally deployed remote weapon system — a high-volume, multi-variant product line

Objective: The Protector RWS family — ranging from RS4 for light vehicles to RS6 with 30mm cannon for armoured platforms to RT20 turret — is one of KDA's most commercially successful product lines, deployed on vehicles in dozens of countries. It is a mechatronics product combining optical sensors, stabilisation systems, fire control computers, and weapons — and managing the variant proliferation across customer-specific vehicle integrations is a live systems engineering challenge.

Engineers who understand this family and can discuss the integration challenges of mounting a stabilised weapon system on a moving vehicle are demonstrating domain relevance. 3 h

Done Signal: You can explain what a remote weapon station is and what operational problem it solves (removing crew from exposed gunner positions), describe the Protector RS4 at a functional level (sensors, stabilisation, fire control, operator interface), explain what a vehicle integration challenge looks like (centre of gravity, electrical interface, vibration environment, certification), and name three current customer nations.

Senior Level *(Principal Engineer, Technical Lead, Fellow)*

All mid-level tasks still apply. NSM/JSM knowledge (task 7) must be operational — you should have worked on guidance, seekers, or propulsion at a technical depth that supports a peer discussion. NASAMS knowledge (task 8) must extend to the fire control architecture, not just the system description.

14. Prepare to Walk Through a Complete Missile or Air Defence System Development Lifecycle — from Operational Requirements Through Qualification Firing 4–6 h

Skill: End-to-end system development ownership for a KDA-class product

Objective: Senior KDA engineers own technical threads that span from operational requirements through system qualification — a process involving extensive simulation, hardware-in-the-loop testing, integration testing, and ultimately live firing at ranges such as Andøya. The interview at this level probes whether you have done this on a real system — not contributed to it. The specific Kongsberg dimension is the export element: qualification for Norway is necessary but not sufficient — each export nation may require additional qualification activities, interface adaptations, or certification evidence.

Done Signal: For one missile, air defence subsystem, or fire control system you personally owned the development thread for, you can answer without notes at every phase — the operational requirement source, the key design trade-off made, the qualification test plan, the hardest technical finding during qualification, how it was resolved, and what additional evidence was required for the first export customer.

15. Prepare a Technically Grounded Position on the 3SM Tyrving Supersonic Missile Programme and What It Represents for Norway's Missile Technology Development 4 h

Skill: Next-generation KDA programme technical awareness at senior engineering depth

Objective: The 3SM Tyrving is a supersonic cruise missile for land-attack and anti-ship roles, led by Kongsberg with Diehl, MBDA Deutschland, and Nammo as subcontractors. This is KDA's most technically ambitious new development — moving from the subsonic NSM to a supersonic weapon requires fundamentally different aerodynamic design, propulsion (ramjet rather than turbofan), guidance at higher dynamic pressure, and warhead design for higher-velocity impact. Senior engineers who can engage with what this transition requires technically are demonstrating the forward-looking technical leadership the 3SM programme needs.

Done Signal: You have a prepared 3-minute technical discussion of one specific engineering challenge in developing a supersonic cruise missile — named specifically (ramjet inlet design and shock management, terminal guidance at high dynamic pressure, aerodynamic heating and structural thermal management, or navigation accuracy at supersonic speeds). Your position is technically grounded and connects your engineering background to the challenge.

16. Prepare Specific Examples of Interfacing with NATO Programme Offices or Multinational Customer Technical Teams — Representing Your Company's Technical Position 2 h

Skill: Multinational programme technical authority at the NATO customer interface

Objective: KDA sells to 14 NSM nations, 13 NASAMS nations, and multiple Protector customers. Managing the technical interface with these customers — who range from the US DoD to the Polish Armaments Agency to the Australian Department of Defence — requires the ability to represent Kongsberg's technical position confidently in English in formal programme reviews, to manage customer-specific requirements that diverge from the baseline, and to coordinate between Kongsberg's Norwegian engineering teams and international customer programme offices. Senior engineers who have done this are immediately deployable in KDA's most commercially active relationships.

Done Signal: You have a story where you represented your organisation's technical position in an international programme review conducted in English — a specific customer, a specific technical question or disagreement, the evidence you prepared and presented, and how the question was formally resolved. The story demonstrates both technical authority and cross-cultural professional composure.

17. Prepare to Discuss Kongsberg's Role in European Defence Architecture — NASAMS for Ukraine, JSM for Japan, NSM Export Expansion 3 h

Skill: Strategic programme context awareness at the level of a senior KDA engineering leader

Objective: Kongsberg's defence business has been transformed by European security developments since 2022. NASAMS deliveries to Ukraine (donated by the US and sourced from Kongsberg), Japan's acquisition of JSM for F-35A (fifth follow-on contract signed December 2025), NSM's adoption by Denmark and the UK, and Australian manufacture of NSM through a new factory near Newcastle — all represent a new chapter in Kongsberg's strategic footprint. Senior engineers who can engage with what this expansion means technically — the qualification variants required for different launch platforms, the through-life support challenges of systems deployed in active conflict, the export control implications of US-origin AIM-120 missiles in NASAMS — are demonstrating the strategic engineering perspective that senior KDA leadership needs.

Done Signal: You can discuss two specific aspects of Kongsberg's recent export expansion with technical specificity — for example: what platform integration modifications were required to qualify NSM for the Royal Navy's Type 23 frigates, or what the manufacturing localisation in Australia means for component qualification and supply chain management. Your discussion is grounded in engineering rather than in geopolitical narrative.

18. Build Quantitative Estimation Fluency for Missile or Air Defence System Performance — Without Simulation Tools 4 h

Skill: System-level performance reasoning at senior engineering speed

Objective: Senior KDA engineers are expected to sanity-check performance analysis from suppliers and junior engineers, assess whether a proposed design change affects range or accuracy estimates at first principles, and respond credibly to "does this make sense?" in programme reviews. For a missile programme: estimating range from fuel mass fraction and Isp, estimating CEP degradation from a guidance error budget change, estimating seeker lock-on range from signal-to-noise ratio and target IR signature. For air defence: estimating engagement envelope from missile speed, altitude ceiling, and handover range. None of these requires a full simulation to estimate at order-of-magnitude if domain fundamentals are solid.

Done Signal: You have worked through five system performance estimation problems in your spe-

cific domain (missile GNC, air defence engagement geometry, DP thruster allocation, or equivalent) — stated out loud, assumptions named before numbers — arriving at order-of-magnitude answers with stated uncertainty. You can identify the most load-bearing assumption in each and how you would verify it. 4 h

All Levels

19. Prepare a Specific Answer to “Why Kongsberg — and Why This Specific Business Area?” That Engages with the Company’s Current Strategic Moment 1–2 h

Skill: Institution-specific motivation that reflects genuine engagement with Kongsberg’s 2025–2026 context

Objective: Kongsberg is not a generic defence or maritime employer. It is a Norwegian national champion that has built world-class missile technology, autonomous maritime systems, and space products from a city of 26,000 people in the Norwegian highlands. Candidates who answer “Why Kongsberg?” with generic interest in advanced technology are missing what makes the company’s moment specifically compelling. The expected answer in 2025–2026 engages with one of: the extraordinary relevance of NASAMS and NSM in the current European security environment, the technical frontier of JSM integration with F-35 as the world’s most advanced air-launched weapon, the maritime autonomy challenge of vessels operating without human oversight, or the specific technical programme you would be contributing to.

Done Signal: You have a 90-second answer that names your target Kongsberg business area and one specific active programme or product, explains one technically specific engineering challenge you find genuinely interesting, connects your background to contributing to it, and demonstrates awareness of Kongsberg’s current strategic context — the European defence environment, the export expansion, or the maritime autonomy frontier, whichever is relevant to your target role. The answer cannot be given unchanged to Raytheon, BAE Systems, or any other employer.

20. Map Your Technical Background to a Specific Kongsberg Business Area, Site, and Product Line — and Verify Security Clearance Eligibility and Norwegian Language Level 1–2 h

Skill: Precise self-positioning within a four-business-area, multi-site organisation

Objective: Kongsberg Gruppen is four substantially different organisations sharing a corporate identity. The engineer targeting KDA’s missile guidance team in Kongsberg city is in a different technical and cultural environment from the engineer targeting Kongsberg Maritime’s DP systems team in Horten or the Kongsberg Digital team in Oslo. A candidate who cannot state their target business area, site, primary product line, current clearance status, and Norwegian language level in 60 seconds is requiring the recruiter to do analytical work they should be doing themselves — and signalling they have not thought carefully about the fit.

Done Signal: You can state in under 60 seconds: the Kongsberg business area you are targeting, the specific site and its primary engineering focus, the product family most relevant to your background, your Norwegian language level (or English-only if targeting an international subsidiary), your clearance eligibility (Norwegian citizenship / EEA / non-EEA), and two specific technical contributions your background makes to the target role. You have verified all of this against the specific vacancy you are preparing for.

SUMMARY REFERENCE TABLE

#	Task Description	Target Level	Effort
1	Apply to summer project track — research specific project first	All	2–3 h
2	Cover letter naming specific product and technical connection	All	2 h
3	ONE KONGSBERG structure — four business areas and cross-links	All	3 h
4	STAR stories — collaboration, quality, cross-discipline, failure	All	2–3 h
5	Norwegian language self-assessment and improvement plan	All	1 h + ongoing
6	Security clearance eligibility — Norwegian system	All	1–2 h
7	NSM and JSM — seeker, guidance, stealth, export reach	Mid+	4–5 h
8	NASAMS — architecture, FDC, AMRAAM integration, NATO IAMD	Mid+	4 h
9	V&V story — safety-critical system, specific failure mode caught	Mid+	2 h
10	Norwegian defence organisation — Forsvarsmateriell and FMA	Mid+	3–4 h
11	Dynamic positioning — how DP works, DP classes, Kongsberg role	Mid+	4 h
12	Cross-organisational technical interface story	Mid+	1–2 h
13	Protector RWS — variants, integration challenges, customers	Mid+	3 h
14	Missile/air defence development lifecycle — to qualification	Senior	4–6 h
15	3SM Tyrfinn — supersonic missile technical challenge position	Senior	4 h
16	NATO/multinational customer technical interface story	Senior	2 h
17	Kongsberg's strategic expansion — NASAMS, JSM, NSM export	Senior	3 h
18	System performance estimation — missile or maritime domain	Senior	4 h
19	"Why Kongsberg — why this business area?" — current context	All	1–2 h
20	Business area/site/product/clearance/Norwegian self-positioning	All	1–2 h

THE STRUCTURAL REALITY

Kongsberg Gruppen is headquartered in a Norwegian city of 26,000 people and employs roughly 13,000 engineers worldwide. It is the company that built the missile system now protecting Kyiv from Russian air attack. It is the company that integrated the most capable air-launched cruise missile in history into the F-35's internal weapons bay. It has put dynamic positioning systems on more than 30,000 vessels. It has launched satellites, built autonomous underwater vehicles, and written the software that guides ships through the most demanding offshore environments on earth.

All of this from Norway — a country of five million people that has consistently punched far above its industrial weight in defence and maritime technology by combining deep engineering specialisation with an unusual willingness to co-develop and co-produce with international partners, and by maintaining majority state ownership of the company since its founding in 1814.

The engineers who succeed at Kongsberg are those who find genuine satisfaction in building things that matter in the most operationally consequential environments that exist — systems that must work the first time, in sea-skimming flight at wave height, against defended targets, at temperatures from Arctic Norway to the Australian littoral. The ones who struggle are those who underestimate how seriously Kongsberg takes the precision of what it does — and how directly that precision is reflected in every engineering decision the company makes.

————— BREAK INTO AEROSPACE —————